Which Acceleration Does Inertia Listen To? The Ignatiev High-Latitude Window as an Intra-Modified-Inertia Discriminator, and a Pre-Registered Exact Null for de Sitter–Unruh Modified Inertia

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Prediction and discriminator note. No data are presented and no detection is claimed. All numerical claims in this note are generated by the committed script `shlem_discriminator.py` (this repository), which runs from first principles and exits 0; every number quoted below matches its output.

Abstract

Ignatiev (2007, 2008) showed that if modified-inertia (MI) MOND couples to a body's *coordinate* acceleration in the Galactic frame, then twice a year, near the equinoxes, at latitude $\approx 80°$ N/S, the Earth's spin-centripetal and orbital accelerations momentarily cancel, dropping a bench-mounted body's kinematic acceleration below a_0 for a fraction of a millisecond and producing a spontaneous-displacement blip — his full interferometer treatment gives $\sim 10^{-14}$ m (the SHLEM effect). The effect is entirely his; this note asks what it discriminates *within* the MI class. We reproduce his window from the kinematic budget (cancellation latitude 79.92° against his derived $79°50' = 79.83°$; window duration 0.43 ms at $a_0 = 9.354 \times 10^{-11}$ m s^{-2}, 0.53 ms at the alternative footing 1.136×10^{-10}), and we reproduce his $\sim 10^{-14}$ m amplitude via the suspension-response bridge $\delta x \approx a_0\,\Delta t/\omega_0$ (9.9×10^{-15} m canonical / 1.5×10^{-14} m alt at his LIGO $\omega_0 = 4.1$ s^{-1}); the naive rigid-body in-window floor is 8.8×10^{-18} m canonical / 1.6×10^{-17} m alt. The window is robust to frame choice: Ignatiev's cancellation budget explicitly includes the Sun's galactocentric acceleration as a vector, and his continuity theorem guarantees at least two exact zeros per year. We then show that the de Sitter–Unruh MI framework — in which inertia is a response to the dS–Unruh bath at the body's *proper* acceleration, with a committed memory kernel $\tau_{\text{mem}} = 2Z/H_\Lambda = 203$ Gyr — predicts an *exact* null in the SHLEM transient channel, by two independent routes: (A) a supported bench mass has proper acceleration $g \approx 9.8$ m s^{-2} at all times, unchanged by the kinematic cancellation, so the time-tagged transient is zero exactly; (B) even if the kinematic variable were (wrongly) fed into the framework's kernel, the 203-Gyr memory averages the millisecond window down by a duty factor 6.7×10^{-23} (canonical; 8.2×10^{-23} alt), to at most $\sim 7 \times 10^{-37}$ m ($\sim 8 \times 10^{-37}$ m alt). The experiment is therefore a kill-condition-only observable for this framework: a blip at Ignatiev's magnitude and timing falsifies both the proper-acceleration coupling and the long-memory kernel simultaneously; a null kills short-memory

Galactic-frame kinematic MI while leaving the framework consistent but *not* confirmed, since the null is shared with modified gravity (whose external-field effect also predicts null on Earth) and with plain Newton. The discriminator tests which *variable* inertia couples to — not the value of a_0 or Z, which remain posited throughout. The experiment, proposed in 2007, has never been performed; it requires a LIGO-class displacement sensor ($\sim 10^{-19}$ m/$\sqrt{\text{Hz}}$) at $\sim 80°$ latitude (Svalbard 78° N; Antarctic Dome A 80.4° S).

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1. Introduction

MOND phenomenology (Milgrom 1983) admits two broad dynamical realizations: modified gravity (MG), in which the field equation sourced by matter is altered, and modified inertia (MI), in which the relation between force and motion is altered below the acceleration scale $a_0 \approx 10^{-10}$ m s^{-2}. Within the MI class a question arises that has no analogue in MG: **which acceleration is the theory's variable?** Candidates include the body's coordinate (kinematic) acceleration in some preferred frame, its proper acceleration (what an on-board accelerometer reads), or a functional of its full trajectory — Milgrom (1994) showed that any MI theory respecting Galilean invariance and the MOND limits must in fact be *nonlocal in time*, so a kernel or memory scale is part of the specification, not an optional extra.

Ignatiev (2007, PRL 98, 101101; 2008, PRD 77, 102001) discovered what he argued is the one terrestrial situation where this question becomes experimentally accessible. His observation, which we state with full credit — **the effect is his** — is that for a kinematic reading of MI, the Earth itself briefly manufactures a deep-MOND laboratory. A bench-mounted body at latitude λ carries a spin-centripetal acceleration $\omega^2 R_\oplus \cos\lambda$ and shares the Earth's orbital centripetal acceleration around the Sun. Twice a year, near the equinoxes, at the latitude where these magnitudes match, the acceleration vectors in his cancellation budget momentarily sum to zero and the body's kinematic acceleration in the Galactic frame sweeps through zero, dipping below a_0 for a fraction of a millisecond. If inertia couples to that kinematic variable with an effectively short memory, the body's response to the ambient residual force is momentarily anomalous, producing a spontaneous-displacement transient — the Static High-Latitude Equinox Modified-inertia (SHLEM) effect, his coinage. Ignatiev's full treatment, with a suspended-mirror interferometer response, predicts a signal of order 10^{-14} m.

This note makes one point: the SHLEM channel is not merely a MOND-versus-Newton test — it is a discriminator *within* the MI class, separating theories by the variable and the memory they assign to inertia. We work it through for a specific committed framework: de Sitter–Unruh modified inertia, in which inertia is the response to the de Sitter–Unruh bath at the body's **proper** acceleration,

$$T(a) \;=\; \frac{\hbar}{2\pi k_B c}\sqrt{a^2 + (cH_\Lambda)^2},$$

with acceleration scale $a_0 = cH_\Lambda/Z$, $Z = \sqrt{32\pi/3} = 5.78881$ (posited, not derived), and a committed memory kernel $\tau_{\rm mem} = 2Z/H_\Lambda = 203$ Gyr (Zimmerman 2026a,b). Throughout we carry both numerical footings of a_0: the canonical pure-Λ value $a_0 = c^2\sqrt{\Lambda/32\pi} = 9.354 \times 10^{-11}$ m s^{-2} and the alternative $a_0 = cH_0/Z = 1.136 \times 10^{-10}$ m s^{-2}. The interpolating kernel this framework employs, $\nu(y) = \sqrt{1 + 1/y}$, is identical in form to Milgrom (1999, PLA 253, 273, Eq. 9); the framework's distinctive content is the coefficient (cH_Λ/Z rather than $2cH_\Lambda$) and the MI completion, and we credit that wellhead explicitly.

The punchline is stark. For this framework the SHLEM transient is **zero exactly** — not small, not suppressed: zero — and it is zero by two logically independent routes. That makes SHLEM a genuine dual kill switch for the framework, and this note is written partly to hand that falsification route to its critics in pre-registered form.

2. The window, reproduced

We first reproduce Ignatiev's window from the kinematic budget, independently of his papers. The relevant accelerations of a bench-mounted mass (script, Section 1):

Term	Magnitude (m s^{-2})
Spin centripetal, equator: $\omega^2 R_\oplus$	3.388×10^{-2}
Earth orbital centripetal: $(2\pi/\mathrm{yr})^2$ AU	5.931×10^{-3}
Galactocentric centripetal: $V_{\rm gal}^2/R_{\rm gal}$	2.146×10^{-10}

with $\omega = 2\pi/86164$ s (sidereal), $R_\oplus = 6.371 \times 10^6$ m, $V_{\rm gal} = 233$ km s^{-1}, $R_{\rm gal} = 8.2$ kpc.

Cancellation latitude. The spin term scales as $\cos\lambda$; it matches the orbital term where

$$\cos\lambda_* = \frac{a_{\rm orb}}{\omega^2 R_\oplus} \;\Rightarrow\; \lambda_* = 79.92°,$$

against Ignatiev's derived $\varphi \simeq \pm 79°50' = 79.83°$ (his PRD, Eq. 3.5 with $|a_s(t_p)| \simeq 0.00593$ m s^{-2}) — agreement to $\sim 0.1°$, the residual traceable to his fuller budget $a_s = a_1 + a_2$ and adopted constants.

Window duration. The residual sweeps at the rate set by the Earth's rotation carrying the spin vector across the orbital one, $\dot{a} \simeq \omega\, a_{\rm orb} = 4.32 \times 10^{-7}$ m s^{-3}. The kinematic acceleration satisfies $|a| < a_0$ for

$$\Delta t = \frac{2a_0}{\dot{a}} = 0.43\,\text{ms}\;\;(\text{canonical}\;a_0 = 9.354{\times}10^{-11})\,, \qquad 0.53\,\text{ms}\;\;(\text{alt}\;a_0 = 1.136{\times}10^{-10}).$$

Ignatiev's PRD does not quote a window duration in these terms; the validation of our Δt is instead the suspension bridge of Section 3, which converts it into his published amplitude.

Frame robustness, and the fork Ignatiev actually states. A tempting shortcut objection runs: the Sun's galactocentric acceleration is 2.146×10^{-10} m s^{-2} — $2.3\,a_0$ (canonical) or $1.9\,a_0$ (alt) — so it "floors the budget above a_0" and no window exists. That scalar-magnitude reading is wrong, and Ignatiev's own construction shows why. He works in the Galactic reference frame S_0 (origin at the Galaxy's center of mass, axes to distant quasars), and his cancellation budget *explicitly includes* the Sun's galactocentric acceleration: his condition is $a_s(t_p) + \omega \times (\omega \times r_1) \approx 0$ with $a_s = a_1 + a_2$, where a_2 is precisely that galactocentric term (his Eq. 3.2). The cancellation is **vectorial**, and the event time t_p is a free parameter: the spin-plus-orbital resultant sweeps through the sky and cancels the galactocentric vector along with everything else. His continuity argument makes this an existence theorem — the parallel component $a_{s\parallel}$ is positive at the summer solstice and negative at the winter solstice, so it has at least two exact zeros per year, and the off-equinox shift is bounded at less than a few days. Nor does the choice of frame matter at the level that could threaten the window: moving to the Local Group frame changes the budget by less than 10^{-12} m s^{-2} (Andromeda's pull; his footnote 2). The window is robust to frame choice.

The frame fork that *does* carry physical content is the one Ignatiev himself states: is a_0 referred to the Galactic (inertial) frame, or is it *invariant* — i.e., does the MOND regime trigger on small acceleration relative to the laboratory? He shows the invariant-a_0 version is internally inconsistent: the kinematic rules of acceleration addition force $\mu(z) \equiv 1$ (his Eqs. 2.1–2.3), and he notes it is ruled out experimentally in any case (Section 6 below). Only the Galactic-frame kinematic reading survives as a coherent theory, and it is the one with the window. Everything below concerns that reading on the kinematic side; the de Sitter–Unruh framework's prediction is null regardless, as Section 4 shows.

3. The kinematic-MI blip

What does a short-memory kinematic MI predict inside the window? Three numbers, all of which we quote.

Naive rigid-body floor. During the window the residual specific force is of order a_0 and the body's inertial response is anomalous; the crudest displacement estimate is ballistic over the window,

$$\delta x_{\text{floor}} \sim \tfrac{1}{2}\, a_0\, \Delta t^2 = 8.8 \times 10^{-18}\ \text{m}\ \ (\text{canonical})\,, \qquad 1.6 \times 10^{-17}\ \text{m}\ \ (\text{alt}).$$

This is the *in-window* displacement of a rigidly mounted mass — a deliberately conservative floor, and **not** what an interferometer records.

The suspension bridge to Ignatiev's amplitude. His 2008 analysis treats the realistic observable: a suspended interferometer mirror. The mechanism, extracted from his Secs. V–VI, is one line — the deep-window **velocity impulse** $\Delta v \approx a_0\, \Delta t$ is displayed by the suspension as a displacement

$$\delta x \ \approx\ \frac{a_0\, \Delta t}{\omega_0} = 9.9 \times 10^{-15}\ \text{m (canonical)}\,, \quad 1.5 \times 10^{-14}\ \text{m (alt)},$$

using his stated LIGO suspension frequency $\omega_0 = 4.1\ \text{s}^{-1}$ ($f_0 = 0.65$ Hz). This reproduces his published amplitude — his Fig. 1 gives 0.95×10^{-14} m at $a_0 = 10^{-10}$, 1.4×10^{-14} at 1.2×10^{-10}, and 3.8×10^{-14} at 2×10^{-10}, the maximum reached $\simeq 0.76$ s after t_p — and explains his a_0^2 scaling, since Δt is itself proportional to a_0. Scaling his Fig. 1 value by $(a_0/10^{-10})^2$ gives per-footing blip targets of 0.83×10^{-14} m (canonical) and 1.23×10^{-14} m (alt). Two caveats travel with these targets: his amplitude is computed with *his* interpolating function $\mu(z) = z/\sqrt{1+z^2}$ and LIGO's ω_0, so the blip-side target is μ- and suspension-dependent at $O(1)$; and the bridge is an order-of-magnitude reconstruction of his numerical solution, not a replacement for it. We take **his** treatment as the kinematic-MI target signal, with the rigid floor as the pessimistic minimum. Either is within reach of LIGO-class displacement metrology ($\sim 10^{-19}\ \text{m}/\sqrt{\text{Hz}}$), given a sensor at the right latitude; no such instrument has ever been sited there (Section 7).

4. The framework's exact null — two independent routes

Route A: the proper-acceleration variable does not blink

The de Sitter–Unruh framework's inertia variable is the body's **proper** acceleration — the argument of the Unruh bath temperature $T(a)$. A bench-mounted mass is *supported*: the bench pushes up with mg, and the mass's proper acceleration is $g \approx 9.8\ \text{m s}^{-2}$ (at the 79.9° site, $g = 9.83\ \text{m s}^{-2}$ — polar gravity, reduced centrifugal; the standard 9.81 shifts the offset below only in its third digit), continuously, through the equinox, through the window, at every instant. The kinematic cancellation is a statement about coordinate acceleration in the Galactic frame; the support force does not blink, the accelerometer reading does not blink, and the bath temperature $T(g)$ does not blink.

One anticipated objection deserves an explicit paragraph. In the framework's galactic phenomenology, a star on a circular orbit is in *free fall* — its proper acceleration is nearly zero — yet the rotation-curve sector evaluates ν at $y =$

$g_{\rm bar}/a_0$, an orbital (relational) acceleration. Does that not concede that the free-fall sector uses a kinematic variable, which the SHLEM window would then trigger? No, because the framework's variable assignment is worldline-class-dependent, and the two sectors never overlap here. For *freely falling* bodies the framework's committed treatment (Zimmerman 2026c) evaluates the bath response on the relational acceleration of the orbit, with the shared galactic free fall self-cancelling; for *supported* bodies the variable is the accelerometer-read proper acceleration. The bench mass **never enters free fall during the window**: the support force is continuous through t_p, so the free-fall rule is never invoked. Only the supported-worldline rule applies, and its variable — proper acceleration $= g$ — is constant. The kinematic near-zero is a coincidence of coordinate accelerations to which no framework input is sensitive.

The SHLEM observable is the *time-tagged transient* correlated with the predicted window. Every variable the framework's inertia responds to is constant through the window, so

$$\boxed{\delta x_{\rm transient} = 0 \text{ exactly — not suppressed: zero.}}$$

There *is* a framework effect on a supported mass, but it is static: with $\nu(y) = \sqrt{1 + 1/y}$ and $y = g/a_0$, the fractional inertia deviation is $a_0/2g = 4.77 \times 10^{-12}$ (canonical; $y = 1.049 \times 10^{11}$) or 5.79×10^{-12} (alt; $y = 8.639 \times 10^{10}$). This offset is present at all times, carries no time tag, and is absorbed into any calibration of the apparatus. It contributes nothing to the transient channel.

Route B: the committed memory kernel freezes millisecond windows

Route A could be contested by someone who insists the kinematic variable is the right input even to this framework's kernel. So we compute that (wrong-on-its-own-terms) hybrid too. The framework's memory kernel is committed, not adjustable: $\tau_{\rm mem} = 2Z/H_\Lambda$ with $H_\Lambda = c\sqrt{\Lambda/3} = 1.806 \times 10^{-18}$ s^{-1}, giving

$$\tau_{\rm mem} = 6.41 \times 10^{18} \text{ s} = 203 \text{ Gyr}$$

(Zimmerman 2026b, where this kernel is load-bearing for the cluster σ-spread prediction). A millisecond excursion of the input variable is averaged over that memory with duty factor

$$\frac{\Delta t}{\tau_{\rm mem}} = \frac{4.3 \times 10^{-4}}{6.4 \times 10^{18}} = 6.7{\times}10^{-23} \ (\text{canonical})\,, \qquad \frac{5.3 \times 10^{-4}}{6.4 \times 10^{18}} = 8.2{\times}10^{-23} \ (\text{alt}),$$

so even Ignatiev's full 10^{-14} m signal would be frozen to at most $\sim 7 \times 10^{-37}$ m (8×10^{-37} m alt) — about 17 orders below a LIGO-class noise floor per root-hertz, i.e., nothing. And the linear duty-factor estimate is itself a *conservative* *upper bound*: the kernel smooths the **input**, so the kernel-averaged kinematic acceleration moves fractionally by only $\sim \Delta t/\tau \approx 7 \times 10^{-23}$ and never leaves the Newtonian regime even instantaneously — the hybrid never enters deep MOND at all. The framework's SHLEM prediction is dead twice over, by two independent commitments: the variable (Route A) and the kernel (Route B). Falsifying the null therefore falsifies both at once.

5. The discriminator and the pre-registered verdicts

SHLEM partitions the theory space as follows:

Theory class	Variable / mechanism	SHLEM prediction
Short-memory kinematic MI (Ignatiev's target class)	coordinate accel. in the Galactic frame, memory $\lesssim$ window	**Blip**: target 0.83×10^{-14} m canonical / 1.23×10^{-14} m alt (μ- and suspension-dependent at $O(1)$); rigid floor 8.8×10^{-18} / 1.6×10^{-17} m
Proper-acceleration MI (this framework)	proper accel., dS–Unruh bath	**Null, exactly** (Route A; independently Route B)
Long-memory kinematic MI (e.g., any Milgrom-1994-compliant kernel with $\tau \gg$ ms)	trajectory functional	**Null** (window averaged away)
Modified gravity + EFE (AQUAL/QUMOND/AeST)	field equations; Earth is deep in the Sun's + Galaxy's external field	**Null** (EFE keeps local dynamics Newtonian)
Newton/GR	—	**Null**
Quantized inertia (McCulloch)	Unruh-horizon mechanism, but predicting lab-scale anomalies	distinct program; see Section 6

The pre-registered verdicts, both ways:

If a SHLEM blip is seen at Ignatiev's magnitude and timing: the de Sitter–Unruh framework's proper-acceleration coupling *and* its 203-Gyr kernel are both falsified. There is no parameter to move: the variable is the framework's founding postulate and the kernel is committed in print. This is a genuine dual kill switch, and we state it as such — this note hands the framework's opponents a working falsification route.

If SHLEM is null: short-memory kinematic MI is killed on Ignatiev's Galactic-frame reading — which, by his own consistency argument, is the only coherent kinematic reading (the invariant-a_0 alternative is internally inconsistent and already excluded by Gundlach-class laboratory tests; Sections 2 and 6). The null is therefore chargeable, cleanly, to the entire short-memory kinematic class. The de Sitter–Unruh framework is **consistent but not confirmed**: the null is shared with long-memory kinematic MI, with modified gravity (whose external-field effect also predicts a terrestrial null), and with plain Newton. A null moves no posterior weight *toward* this framework relative to those alternatives.

What SHLEM does not test: the value of a_0, or Z. Both remain posited. The window duration depends weakly on a_0 (0.43 vs 0.53 ms across the two footings) and the blip target scales as a_0^2, but a detection or null at either footing carries the same discriminating content. SHLEM tests the **variable** inertia couples to and the **memory** it carries — nothing else.

The discriminator's reach, stated precisely: it separates {short-memory kinematic MI} from {proper-acceleration MI, long-memory kinematic MI, MG, Newton}. It does not separate the members of the second set from one another.

6. Consistency with existing laboratory nulls, and the McCulloch distinction

Gundlach et al. (2007). The most sensitive direct test of $F = ma$ at small accelerations (PRL 98, 150801) verified Newtonian response down to lateral accelerations of 5×10^{-14} m s^{-2}. This experiment does **not** discriminate between the two MI readings under test here, because both pass it:

- *This framework:* the torsion-pendulum test masses are supported; their proper acceleration is $\sim g$, so $y \sim 10^{11}$ and the predicted deviation from Newton is the static $\sim 5 \times 10^{-12}$ fraction of Section 4 — Newtonian to experimental purposes. Pass.
- *Galactic-frame kinematic MI:* at Seattle's latitude the spin-centripetal acceleration alone is $\omega^2 R_\oplus \cos(47.6°) = 2.28 \times 10^{-2}$ m s$^{-2} \gg a_0$, so outside the high-latitude equinox windows the kinematic variable never approaches a_0 either. Pass.

What Gundlach-class tests *do* kill is the third reading: invariant (lab-frame) a_0 kinematic MI, which predicts MOND behavior whenever laboratory acceleration is small — Ignatiev cites exactly this class of result in ruling it out, alongside his internal-consistency argument. The division of labor is clean: Gundlach kills invariant-a_0 kinematic MI; SHLEM tests Galactic-frame kinematic MI; neither touches proper-acceleration MI outside the window. Only the SHLEM window separates the two surviving readings; that is exactly what makes it worth an instrument.

Quantized inertia. McCulloch's quantized inertia (e.g., EPL 90, 29001, 2010) is also Unruh-based, so we mark the distinction once and plainly. QI modifies inertia via a Rindler-horizon Casimir-like cutoff on Unruh modes and has been argued by its author to predict laboratory-scale anomalies (thruster-type effects and related claims). The de Sitter–Unruh framework treated here makes the *opposite* generic prediction: because its variable is proper acceleration and every laboratory body sits at $y = a/a_0 \sim 10^{11}$, it predicts **lab nulls across the board** — including, as computed here, an exact null in the one lab channel (SHLEM) where a kinematic Unruh-flavored MI would predict a signal. SHLEM is one of the channels that empirically separates the two programs.

7. Feasibility and honest scope

Sites, timing, and sensitivity. The window requires latitude near 79.9°; per Ignatiev the latitude prediction is robust — lunar perturbation shifts it by no more than $\sim 6'$ ($\sim$ 10 km) — while the *longitude* of the event spot varies from year to year (his Table II tabulates the 2008–2019 spots) and the event instant is shifted from the exact equinox by less than a few days, all computable years in advance from ephemerides. Realistic sites: Svalbard (Longyearbyen, 78° N — infrastructure, access) and the Antarctic high plateau (Dome A, 80.4° S — seismic quiet, brutal logistics). One instrument-design constraint he derives is worth repeating: at 80° latitude the SHLEM spot runs westward across the surface at $\approx$ 80 m/s, so a reference mirror farther than 80 m from the test mirror is unaffected over the $\sim$ 1 s observation interval; closer, both mirrors' responses must be differenced. The target signal ($\sim 10^{-14}$ m, peaking $\simeq 0.76$ s after t_p, twice a year at a *predicted* UTC instant) is large against a LIGO-class displacement noise floor of $\sim 10^{-19}$ m/$\sqrt{\text{Hz}}$; even the pessimistic rigid floor (8.8×10^{-18} m canonical / 1.6×10^{-17} m alt) is above it in a matched band, with far less margin. The time-tagged nature of the prediction is the experiment's greatest systematic asset: everything uncorrelated with the predicted instant is background. The measurement was proposed in 2007 and, to our knowledge, has never been performed; no displacement sensor of the required class has ever operated at $\sim$ 80° latitude.

Novelty coverage caveat. A literature search found no prior publication of this specific *intra-MI discriminator* reading of SHLEM — Ignatiev's own papers predict the blip for MI generally, without the proper-acceleration/kinematic fork or the memory-kernel route. We state this as "no duplicate found," not "none exists."

Scope, restated without hedging. This note presents no data and claims no detection. It contributes: (i) an independent reproduction of Ignatiev's window (latitude to $\sim 0.1°$, both a_0 footings) and of his $\sim 10^{-14}$ m amplitude via the suspension bridge; (ii) the corrected frame analysis — the window is robust to frame choice, and the operative fork is Galactic-frame versus invariant a_0, with only the former coherent; (iii) a computed, pre-registered *exact zero* for the de Sitter–Unruh MI framework by two independent routes, making SHLEM a dual kill switch for that framework; and (iv) the partition table of Section 5. The framework's $a_0 = cH_\Lambda/Z$ and $Z = \sqrt{32\pi/3}$ are posited inputs here as everywhere; SHLEM cannot test their values and we do not claim otherwise. A null outcome would leave the framework standing in a crowd of theories predicting the same null; only the blip outcome is decisive, and it is decisive *against* the framework. We consider handing over that switch to be the point of the exercise.

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Reproducibility: every dimensional number in this note (accelerations, latitude, window durations, suspension-bridge and Fig.-1-scaled blip targets, displacement floor, static deviations, H_Λ, τ_{mem}, duty factors, suppressed blip) is computed from scratch by `shlem_discriminator.py` in this directory; the script carries both a_0 footings and exits 0.